

Revision of topics from Probability Theory.

Random Experiment (R.E.) (i) all outcomes known in advance  
(ii) outcome of a particular trial not known in advance  
(iii) can be repeated under identical conditions

Sample space  $\Omega$  : set of all possible outcomes of a r.e.  
 $\Omega = \{\omega_1, \omega_2, \dots\}$ , each  $\omega_i$  is an outcome, need not be countable.

If outcome  $\omega$  belongs to a set  $E$ , then event 'E' has happened,  $E \subset \Omega$ .

Ex R.E. : tossing of a coin until H appears.

sample space  $\Omega = \{H, TH, TTH, \dots\}$

Event  $E$  is '# trials required is odd' then

$E = \{H, TTH, TTTTH, \dots\}$ .

Classical definition of probability R.E. has finite no. of equally likely outcomes.

$\Omega = \{\omega_1, \dots, \omega_n\}$ , if  $\omega \in E$ , then outcome  $\omega$  is favourable to event  $E$ ,

and  $\text{Prob.}(E) = \frac{\# \text{ outcomes favourable to } E}{\text{total } \# \text{ outcomes}} = \frac{\# \text{ elements in } E}{\# \text{ elements in } \Omega}$ .

Relative frequency defn of probability R.E. is repeated  $N$  times, out of which event  $E$  happens  $f_N(E)$  times

Then  $\text{Prob.}(E) = \lim_{N \rightarrow \infty} \frac{f_N(E)}{N}$

Both definitions have significant flaws, hence

Axiomatic defn of probability probability space  $(\Omega, \mathcal{F}, P)$ , where

$\Omega$ : sample space

$\mathcal{F}$ :  $\sigma$ -field of subsets of  $\Omega$ , i.e., class of subsets of  $\Omega$  closed under

(i) complementation (ii) union, (iii) containing  $\Omega$ .

(example, for a finite  $\Omega$  with  $n$  elements, the powerset with  $2^n$  elements)

$P$ : probability function/measure, a real valued set fn defined on  $\mathcal{F}$ , with range  $[0, 1]$

(i)  $\forall A \in \mathcal{F}$ ,  $P(A) \geq 0$  (non-negativity)

(ii) For  $A_1, A_2, \dots \in \mathcal{F}$ ,  $A_i \cap A_j = \emptyset \forall i \neq j$ ,  $P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$  ( $\sigma$ -additivity)

(iii)  $P(\Omega) = 1$  (normed).

Random Variable is a  $\tilde{A}$  real valued fn  $X: \Omega \rightarrow \mathbb{R}$ , defined on  $\Omega$ . ~~is a random variable~~

More rigorously,  $X^{-1}(B) = \{\omega: X(\omega) \in B\} \in \mathcal{F} \quad \forall B \in \mathcal{B}$ .

where  $\mathcal{B}$ : Borel sets, subsets of  $\mathbb{R}$

$\mathcal{B}$ :  $\sigma$ -field of subsets of  $\mathbb{R}$  or Borel  $\sigma$ -field,

i.e.,  $X$  is a r.v. if the inverse images of all Borel sets in  $\mathbb{R}$  are events.

In practice, need not consider all Borel sets  $B \in \mathcal{B}$ , but a class of subsets of  $\mathbb{R}$  that generates  $\mathcal{B}$ , e.g.,  $(-\infty, x)$  or  $(-\infty, x]$ ,  $\forall x \in \mathbb{R}$ .

i.e.,  $X$  is a r.v. if  $\forall x \in \mathbb{R}$ ,  $X^{-1}((-\infty, x]) = \{\omega: X(\omega) \leq x\} \in \mathcal{F}$ .

Example. R.E. a coin is tossed twice

$\Omega = \{HH, HT, TH, TT\}$ ,  $\mathcal{F}$ :  $\sigma$ -field of subsets of  $\Omega$ .

$X = \#$  heads in  $\omega$  (an outcome).

Is  $X$  a r.v.?

Yes,  $X(\omega) = \begin{cases} 0 & \text{if } \omega = TT \\ 1 & \text{if } \omega = TH, HT \\ 2 & \text{if } \omega = HH \end{cases}$

and  $X^{-1}((-\infty, x]) = \begin{cases} \emptyset & \text{if } x < 0 \\ TT & 0 \leq x < 1 \\ TT, HT, TH & 1 \leq x < 2 \\ \Omega & x \geq 2 \end{cases}$

$\because$  each of  $\emptyset, TT, TT, HT, TH, \Omega \in \mathcal{F}$   
hence  $X$  is a r.v.

Common notation  $X$  is a r.v. on prob. space  $(\Omega, \mathcal{F}, P)$ , and  $(\mathbb{R}, \mathcal{B}, P_X)$  is the prob. space induced by  $X$ .

$$(\Omega, \mathcal{F}, P) \xrightarrow{X} (\mathbb{R}, \mathcal{B}, P_X)$$

Distribution function of a random variable.

A function  $F_X: \mathbb{R} \rightarrow [0, 1]$  s.t.

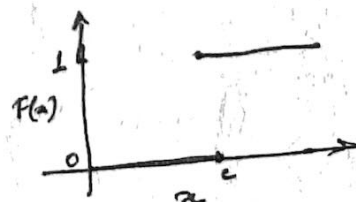
$$F_X(x) = P_X[(-\infty, x]] = P(\omega: X(\omega) \leq x), \text{ then}$$

$F_X(\cdot)$  is the cumulative distribution fn (cdf) or joint dist. fn of the r.v.  $X$ .

Ex 1.  $X(\omega) = c \quad \forall \omega \in \Omega$ , so that

$$P_X(X=c) = P(\omega: X(\omega)=c) = P(\Omega) = 1. \text{ Then}$$

$$F_X(x) = P_X(X \leq x) = P(\omega: X(\omega) \leq x) = \begin{cases} 0 & \text{if } x < c \\ 1 & \text{if } x \geq c \end{cases}$$



Note,  $F$  is non decreasing, right continuous, but not left continuous  $\forall x \in \mathbb{R}$ ,  $F(-\infty) = 0$ ,  $F(\infty) = 1$ .  $F(\cdot)$  has 1 point jump discontinuity at  $c$ .

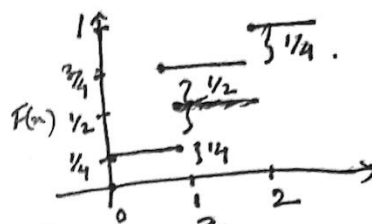
Ex 2. Tossing a fair coin ( $P(H) = P(T) = \frac{1}{2}$ ) twice.

$$\Omega = \{HH, HT, TH, TT\}.$$

$X(\omega) = \# \text{ heads in } \omega$ ,  $X$  is a r.v.

~~NOTE~~  $P(X=0) = \frac{1}{4}$ ,  $P(X=1) = \frac{1}{2}$ ,  $P(X=2) = \frac{1}{4}$ .

Then  $F_X(x) = P(X \leq x) = \begin{cases} 0 & \text{if } x < 0 \\ \frac{1}{4} & 0 \leq x < 1 \\ \frac{3}{4} & 1 \leq x < 2 \\ 1 & x \geq 2 \end{cases}$



$F(\cdot)$  has 3 pts of jump discontinuity at 0, 1, 2.

Ex 3.  $\Omega = [a, b]$ ,  $\forall I \subset \Omega$ ,  $\text{Prob}(I) = \frac{\text{length of } I}{b-a}$

$X(\omega) = \omega$  (a pt. in  $\Omega$ );  $F_X(x) = P(X \leq x) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{b-a} & a \leq x < b \\ 1 & x \geq b. \end{cases}$

$F(\cdot)$  is continuous everywhere.

NOTE If  $F(\cdot)$  is distn fn of a r.v.  $X$  defined on prob. sp.  $(\Omega, \mathcal{F}, P)$ , then  
 (i)  $F$  is non-decreasing (ii)  $F$  is right continuous,  $F(x+) = \lim_{h \downarrow 0} F(x+h) = F(x)$ ,  
 (iii)  $F(\infty) = \lim_{x \rightarrow \infty} F(x) = 1$ , (iv)  $F(-\infty) = \lim_{x \rightarrow -\infty} F(x) = 0$ .

Converse is also true, i.e., if a fn  $F$  satisfies properties (i) - (iv), then it is the distn fn of a suitably defined r.v.  $X$ .

Any distn fn  $F$  is continuous if it is left continuous (besides (i) - (iv)), i.e., at  $a \in \mathbb{R}$ , if  $F(a-) = F(a)$ .

Discrete r.v.  $X$ .  $F_X$  is not left continuous, but can have only countable discontinuities.  $X$  has pmf  $P(X=a) = F(a) - F(a-) > 0$  (since  $F$  non-decreasing) for  $a \in \mathcal{X}$ , range of  $X$   $= 0$ , or.

For  $a < b$ ,  $P(a < X < b) = F(b-) - F(a)$   
 $P(a \leq X \leq b) = F(b) - F(a-)$   
 $P(a < X \leq b) = F(b) - F(a)$   
 $P(a \leq X < b) = F(b-) - F(a-)$

Continuous r.v.  $X$  if  $\exists$  non-negative integrable fn  $f: \mathbb{R} \rightarrow [0, \infty)$   $\Rightarrow$  for any  $x \in \mathbb{R}$ ,  $F(x) = \int_{-\infty}^x f(t) dt$ ,  $f(\cdot)$  is the pdf of

l.e. contn. r.v.  $X$ .

Hence  $P(X=a) = F(a) - F(a-) = 0$  + all 4 above prob.  $\leq 1$  equal to  $F(b) - F(a)$ .

Mixed type r.v.,  $F(x) = \alpha F_d(x) + (1-\alpha) F_c(x)$ ,  $0 < \alpha < 1$   
 $F$  is a mix of a discrete part  $F_d$  and a continuous part  $F_c$ .